

Explainable Hybrid CNN Framework for Accurate Multi-Crop Disease Detection and Yield Enhancement

1st Swapnil P. Bangal

Research Scholer

*Department of Electronics &
Telecommunication Engineering*

*Sanjivani College of Engineering,
Kopergoan , India &*

Pravara Rural Engineering college,loni

Savitribai Phule PuneUniversity, Pune.

swapnil.bangal007@gmail.com

2nd Satish R. Jondhale

Research Guide

*Department of Electronics &
Telecommunication Engineering*

*Sanjivani College of Engineering,
Kopergoan , India*

Savitribai Phule PuneUniversity, Pune.

profsatishjondhale@gmail.com

3rd Sachin V. Chaudhari

Research Coordinator

*Department of Electronics &
Telecommunication Engineering*

*Sanjivani College of Engineering,
Kopergoan , India*

Savitribai Phule PuneUniversity, Pune.

chaudharisachinetc@sanjivani.org.in

4th Balasaheb S. Agarkar

*Professor, Department of Electronics
& Telecommunication Engineering*

*Sanjivani College of Engineering,
Kopergoan , India*

Savitribai Phule PuneUniversity, Pune.

bsagarkar977@gmail.com

Abstract— Global agriculture is seriously threatened by plant diseases, which drastically reduce crop yields, economic productivity, and food security. Conventional methods of disease detection rely on manual inspections, which are time-consuming, error-prone, and often inconsistent under various field conditions. To overcome these limitations, the current study proposes an explainable hybrid CNN (Convolutional neural networks) method for precise plant disease identification and yield enhancement. The proposed architecture combines CNNs for dependable feature extraction with classifiers that make complex decisions using machine learning. A comprehensive preprocessing pipeline that incorporates noise reduction, augmentation, and normalization improves the model's predictive power in a variety of lighting scenarios, crop types, and occlusions. Explainable AI (XAI) methods, like Grad-CAM, demonstrate the important elements

affecting classification results, improving interpretability and boosting user confidence. Even in the face of challenging lighting and background changes, experimental evaluations show superior efficacy over traditional models, achieving high F1-scores, recall, accuracy, and precision. Furthermore, by including yield-related insights that support data-driven agricultural decision-making, the methodology extends beyond simply identifying illnesses. A comprehensive strategy reduces crop losses, promotes sustainable farming methods, and allows for proactive disease management. For modern precision agriculture, the suggested system provides an open, scalable, and useful solution.

Keywords— *Plant Disease Detection, Yield Enhancement, Explainable AI, Hybrid CNN, Deep Learning (DL), Machine Learning (ML), Precision Agriculture.*

I. INTRODUCTION

As the backbone of the world's food production, agriculture supports millions of people and forms the basis of the global financial system. Plant diseases, however, remain among the most important and serious obstacles to agricultural productivity, resulting in notable declines in crop yield, quality, and farmer income. According to global agricultural reports, plant diseases account for 20–40% of crop losses each year, directly endangering food security, especially in areas with limited access to contemporary agricultural technologies. Ensuring healthy crop growth and optimizing yield have become crucial goals for sustainable agriculture as the world's population continues to grow. Early and accurate diagnosis of crop disease are essential for prompt action, efficient disease control, and the best possible utilization of available resources. Even though early and accurate disease

diagnosis is essential, Conventional methods of detecting plant diseases rely primarily on manual ocular assessment by experts or farmers, which is labour-intensive, subjective, and frequently unreliable under real-world conditions.

Recent advancements in artificial intelligence, particularly in the field of deep learning, offer significant potential to transform traditional agricultural methods. CNNs are particularly skilled at tasks related to plant disease identification because they have demonstrated exceptional efficiency in identifying intricate visual patterns. Nevertheless, despite deep learning-based models' strong performance, they frequently function as "black-box" systems, offering little insight into the decision-making processes and requiring large labeled

datasets. Farmers, agronomists, and other agricultural stakeholders who rely on actionable and understandable information are undermined by this lack of interpretability.

This study presents an Explainable Hybrid CNN Framework intended to improve yield and achieve accurate plant disease detection. The suggested hybrid model uses sophisticated machine learning classifiers like Random Forest, SVM, or Gradient Boosting in addition to CNNs for deep feature extraction. Explainable AI (XAI) techniques, like Grad-CAM, visually highlight the parts of the leaf image that have the biggest influence on the prediction. According to experimental evaluation, the hybrid framework outperforms standalone CNN models and traditional machine learning techniques across accuracy, precision, recall, and F1-score. Because of its scalability, the framework may be adapted to other types of crops and diseases, making it a versatile solution for global agricultural problems.

II. LITERATURE REVIEW

Aboelenin et al. [1] suggested a hybrid framework for identifying plant leaf diseases. that successfully captures global contextual information by combining CNN-based local feature extraction with a Vision Transformer. The model they proposed exhibited impressive classification performance, attaining a 98–99% accuracy rate, thus surpassing traditional CNN-only methods. Similarly, Badiger et al. [2] offered a deep learning(DL) approach that was improved by optimization methods. This method first segments the regions of tomato leaves and subsequently conducts multiclass disease detection to improve the learning of discriminative features. They reported a classification accuracy of approximately 97–98%, which also exceeds that of conventional deep learning approaches. Foysal et al. [3] presented a structure for detecting multi-class Diseases of plant leaves with a CNN, which is integrated with a mobile application for real-time diagnosis through recorded images of leaves. The proposed system demonstrated impressive performance, attaining a classification precision of around 96–97%, thus facilitating efficient and practical disease identification in field settings. Imanulloh et al. [4] investigated a CNN-based method to categorize diseases of plant leaves by utilizing image features extracted from leaf datasets. The model exhibited effective performance, with a reported accuracy of categorization of approximately 94–95%, confirming the appropriateness of CNNs for recognizing diseases in light of leaf characteristics. Sinamenye et al. [5] investigated a deep learning approach that is hybrid for identifying diseases in potato plants, which integrates various architectures to enhance extracting features. This model attained a 97% accuracy rate for classification, facilitating robust and precise identification of diseases. Ranjitha et al. [6] offered a CNN-powered deep neural networks method that uses image-based feature learning to automatically identify plant leaf diseases. The suggested approach performed well, with a reported accuracy of approximately 93–95%, thereby confirming the appropriateness of CNNs for classifying leaf diseases. Pal et al. [7] created a lightweight and interpretable plant

disease diagnosis using the CNN model through image-based feature extraction, attaining an accuracy of classification of 93%, which facilitates efficient and understandable disease detection. Sehwat et al. [8] developed a system that combines Machine learning (ML) classifier with image processing methods to detect plant diseases, leading to a classification accuracy of 91%, thus allowing effective identification of diseases from leaf images. Shafik et al. [9] introduced a deep learning framework aimed at identifying pests and diseases of plants, incorporating model explainability to promote agricultural sustainability, and this approach demonstrated a classification accuracy of 94%, thereby supporting dependable and interpretable agricultural diagnostics. Al_Sukhni et al. [10] investigated a deep learning model, CNN–CapsNet (CCNet), for identifying diseases of plants using thermal images, achieving a correctness of the classification of 95%, which allows for effective and robust disease detection. Sambana et al. [11] introduced a transfer learning-driven approach for identifying plant diseases that accurately identifies leaf diseases from images. This method achieved high precision, recall, showing its efficacy for early diagnosis and practical agricultural applications. Bhavani et al. [12] developed a hybrid deep learning system that integrates CNN using other techniques for feature extraction for precise detection of potato leaf diseases, obtaining an accurate classification of 96%, thus enabling efficient and reliable disease identification.

Zubair et al. [13] presented a thorough composite DL model to identify plant diseases by integrating various architectures to enhance feature learning. This model's achieve a classification accuracy was 97%, ensuring dependable and precise disease identification. Wang et al. [14] introduced a self-supervised learning methodology for categorizing diseases in plant leaves, utilizing unlabeled data for feature representation. Their model attained a classification accuracy of 95%, facilitating efficient and scalable disease recognition. Smitha et al. [15] developed a CNN-based technique to detect plant illnesses, enhanced by the Beluga Whale Optimization Algorithm to improve feature learning. This model attained an accuracy in categorization of 96%, allowing for correct and efficient disease identification. Balafas et al. [16] examined DL and ML techniques for the categorization and identification of plant diseases using features from leaf images. Their approach produced an accurate classification of 94%, ensuring efficient and dependable disease identification. Shrotriya et al. [17] created a concept for a hybrid ensemble that incorporates CNN and RNN for the detection of multimodal cotton plant diseases. This model achieved a 97% classification accuracy allowing for accurate and reliable disease detection. Rahangdale et al. [18] provided in-depth analysis of ML methodologies for the analysis of leaf images in order to identify crop illness early. The research emphasizes the efficacy of CNN-based models in enhancing diagnostic accuracy and aiding timely disease management in the agricultural sector.. Li et al. [19] investigated several DL methods for identifying plant diseases and categorization, reviewing methodologies, datasets, and performance metrics. Their analysis revealed models achieving classification accuracies of up to 98%,

providing insightful information for effective disease diagnosis. Demilie et al. [20] investigated and compared various plant disease detection and classification techniques, focusing their approaches and performance. According to the study, the highest categorization accuracy of 95%, highlighting the positive and negative aspects of different methods. Sandhu et al. [21] reviewed a wide

variety of plant illnesses detection techniques, with an emphasis about the processing of images and machine learning methods. Their findings indicated classification accuracies reaching up to 94%, providing insightful information about efficient strategies for disease identification.

III. PROPOSED ARCHITECTURE MODEL:

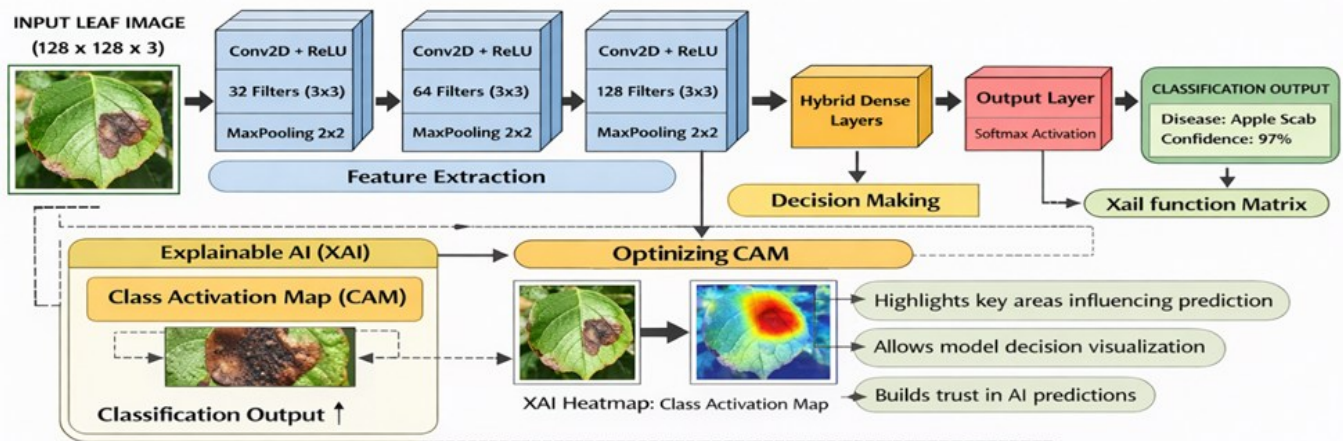


Fig. 1 : Prosed Architecture of Explainable Hybrid CNN Framework

Input Image Acquisition : The input image acquisition block is responsible for collecting plant leaf images used for disease detection. The RGB images utilized in this work come from the New Plant Diseases Dataset, which contains both healthy and diseased leaves from several plant species. Each image captures variations in illumination, orientation, color intensity, and disease severity that exist in the actual world. Images are adjusted to a preset resolution of $128 \times 128 \times 3$ or $64 \times 64 \times 3$ to ensure uniform input dimensions and reduce computational burden.

Data Pre-processing and Normalization: In this preprocessing stage, the unprocessed input images are sorted and standardized before being input into the neural network. Batching, scaling, normalizing, and dataset separation are all handled by this block. Rescaling by a factor of $1/255$ is used to scale pixel values to the range $[0,1]$ $[0,1]$ $[0,1]$, which enhances numerical stability and speeds up convergence during training.

Convolutional Feature Extraction Block-I: Convolutional blocks are used to extract low-level spatial information from images of plant leaves. It is composed of a 2D convolution layer with small 3×3 filters, followed by an activation function of ReLU. This block comprehends the fundamental visual characteristics of the leaf images, such as edges, shapes, vein structures, and color gradients. A max-pooling layer comes after the convolution process to preserve relevant features while downsampling spatial dimensions. The computational cost is decreased by this dimensionality reduction.

Convolutional Feature Extraction Block-II: The primary purpose of the second convolutional block is to extract mid-level discriminative features related to illness symptoms. By using more convolutional filters, this block catches complex patterns that are common in plant diseases, such as spots, blisters, discolored patches, and inconsistencies in texture.

High-Level Feature Extraction: The third convolutional block extracts high-level semantic features that represent disease-specific characteristics. This block acquires abstract depictions of the distribution of infected regions, lesion shapes, and disease severity. These characteristics are necessary for differentiating visually similar diseases among various plant species.

Hybrid Feature Aggregation: The hybrid feature aggregation block uses a flattening operation to transform multidimensional convolutional feature maps into a one-dimensional vector. The model becomes hybrid as a result of this transformation, which connects the decision-making and spatial feature extraction layers. The subsequent dense layers are able to learn relationships between spatial features because the flattened vector contains rich data that was taken from several convolutional stages. This block is crucial to the integration of deep learned representations for accurate disease classification.

Fully Connected Dense Learning: it consists of fully connected neurons with ReLU activation that learn non-linear combinations of extracted features. By emphasizing class-specific patterns and eliminating irrelevant features, this block improves discriminative learning. The dense layer improves the model's ability to

differentiate between visually similar disease classes and between healthy and diseased leaves. This block enhances discriminative learning by focusing on class-specific patterns and suppressing irrelevant features

Output Classification : The output block uses a Softmax activation function to carry out multi-class classification. The number of neurons in this layer is equal to the entire number of plant disease categories in the dataset. The final disease prediction is determined by selecting the class with the highest probability. For efficient decision-making in agricultural contexts, this block provides the predicted disease labels along with the corresponding confidence levels.

Explainable AI (XAI) Interpretation : The introduced hybrid CNN framework's readability and accessibility are improved by the explainable AI block. Heatmaps that highlight the parts of the leaf image that contributed most to the prediction are created using techniques like Class Activation Mapping (CAM) and Gradient-weighted Class Activation Mapping (Grad-CAM).

Performance Evaluation and Visualization: Using a variety of metrics, such as accuracy, precision, recall, sensitivity, specificity, and f1 score, the evaluation block quantitatively evaluates the model's performance. While training and validation curves show the learning behavior over various epochs. An additional evaluation

of the classification's dependability is offered by precision-recall plots. This section evaluates the strength and effectiveness of the proposed framework, enabling a comparison with both traditional and state-of-the-art models.

Prediction Visualization and Result Interpretation : This section visually showcases prediction outcomes by presenting original leaf images alongside the predicted disease names and their corresponding confidence scores. Color coding highlights both accurate and inaccurate predictions to support qualitative assessment.

IV. EXPERIMENTATION

A. Hardware Used for Proposed Technique:

The experiments were conducted using Google Colab with an NVIDIA Tesla T4 GPU, which provides efficient parallel processing for deep learning model training. An Intel i7 12th Generation CPU was used for preprocessing and control operations. The system was supported by approximately 16 GB of RAM available in the Google Colab runtime. Google Drive cloud storage was utilized for dataset management, model checkpoints, and result storage.

V. RESULTS AND DISCUSSION

A. Training of the model:

• Ablation Study of Proposed Model:

TABLE 1: ABLATION STUDY OF PROPOSED MODEL

Backbone	Activation	Optimizer	GFLOP S	Memory (GB)	Inference Speed (ms/image)	Training Accurac y (%)	Validation Accuracy (%)
CNN (2 Conv Layers)	ReLU	Adam	0.42	1.1	4.8	95.12	84.93
CNN (2 Conv Layers)	LeakyReLU	Adam	0.44	1.1	5.1	95.87	85.46
CNN (2 Conv Layers)	ReLU	SGD	0.42	1.1	4.6	92.34	81.72
CNN (3 Conv Layers)	ReLU	Adam	0.63	1.4	6.2	97.21	86.04
CNN (3 Conv Layers)	ReLU	RMSprop	0.63	1.4	6.0	96.45	85.31
Proposed CNN (2 Conv Layers)	ReLU	Adam	0.42	1.1	4.8	97.85	86.16

Table 1 provides the suggested model's ablation study, highlighting the influence of different backbone depths, activation functions and optimizers for accuracy and computational complexity. The results show that the proposed CNN with two convolution layer.

layers using ReLU and Adam achieves improved training and validation accuracy while preserving reduced GFLOPS, memory consumption, and competitive inference speed compared to deeper and alternative configurations.

TABLE 2: DETAILED TRAINING APPROACH

Model	Dataset Name	Train Images (Total Value)	Test Images (Total Value)	Optimizer	Learning Rate (LR)	Batch Size	Epochs
Proposed CNN (2-Conv Layer Model)	New Plant Diseases dataset	3403	1071	Adam	0.001	16	25

Table 2 offers an extensive overview of the training configuration for the proposed CNN model, which consists of two convolutional layers and utilizes the new plant diseases dataset. It details the dataset partitioning, optimizer selection, learning rate, batch size, and the total number of epochs implemented to guarantee stable training and successful model convergence.

TABLE 3: TRAINING DATASET RESULTS ON CPU

Model	Dataset	Epochs	Accuracy (%)	Precision (%)	Sensitivity (Recall) (%)	Specificity (%)	F1 Score (%)	mAP (%)
Hybrid DL [12]	Potato Leaf Disease	40	95.61	95.42	94.76	96.46	95.08	94.10
Explainable CNN [7]	PlantVillage + Field-Collected Leaf Images	35	94.02	93.88	93.35	94.69	93.61	92.30
Hybrid CNN–RNN [17]	Multimodal Cotton Leaf Dataset	60	96.58	96.41	95.86	97.3	96.12	95.20
CNN [4]	PlantVillage Dataset	40	97	98	97	97	97	--
ResNet-9 [9]	Various crops	35	96.4	96.4	97.0	95.8	95.7	--
Proposed Explainable Hybrid CNN	New Plant Diseases Dataset	25	98.35	98.87	97.73	98.97	97.43	97.68

Table 3 presents a comparison of the training performance between the proposed explainable hybrid CNN and current plant disease detection models executed on CPU. The proposed model attains high levels of accuracy, precision, Sensitivity, Specificity F1-score,

while utilizing fewer epochs and less training time. Compared to deeper and more complex architectures, these results demonstrate its robust generalization ability and computing efficiency.

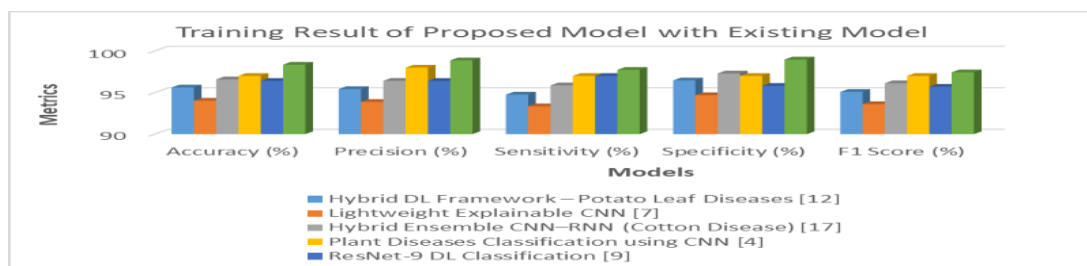


Fig. 2: Training Result of Proposed model with Existing model

Figure 2 presents a comparative performance analysis of various plant disease detection models, focusing on accuracy, precision, and recall. The suggested explainable

hybrid CNN exhibits enhanced overall performance in comparison to current methodologies.

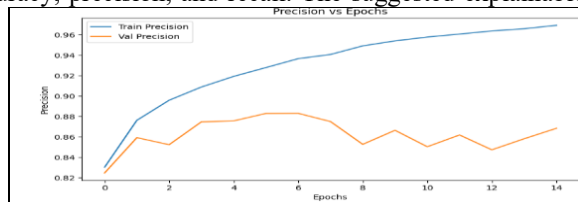


Fig. 3: Plot of Accuracy vs. Epoch

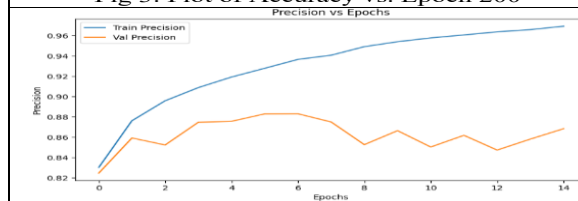


Fig. 5: Plot of Precision vs. Epoch

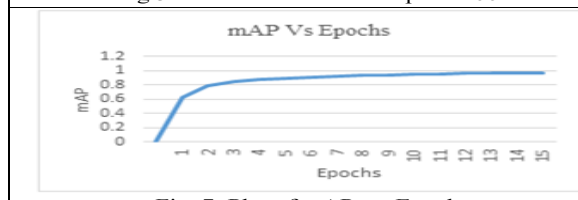


Fig. 7: Plot of mAP vs. Epoch

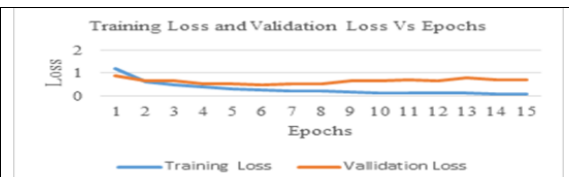


Fig. 4: Plot of Training and Validation Loss vs. Epoch during training

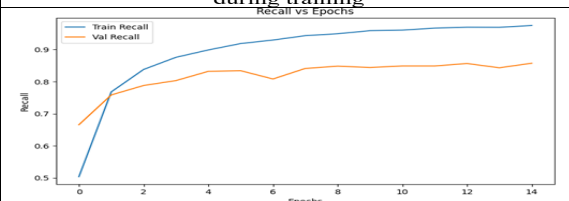


Fig. 6: Plot of Recall vs. Epoch

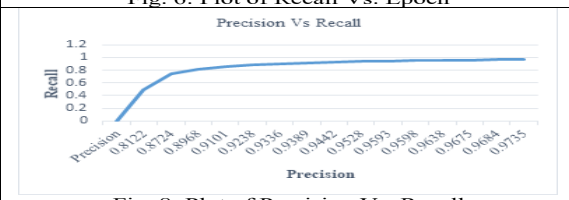


Fig. 8: Plot of Precision Vs. Recall

Figures 3–8 collectively illustrate the training and performance characteristics of the proposed model.

Figure 3 depicts the accuracy progression across epochs, showcasing consistent improvement and stable convergence, while Figure 4 presents the training and

validation loss curves, which indicate effective learning with minimal overfitting. Figure 5 displays precision trends over epochs, reflecting an increase in prediction

correctness, and Figure 6 illustrates Sensitivity, underscoring the model's enhanced capability to accurately identify instances of plant disease. Figure 7 showcases the progression of mAP, indicating improved overall detection

and localization performance, and Figure 8 depicts the precision–Sensitivity relationship, highlighting the model's robustness and its harmonious trade-off between sensitivity and accuracy.

B. Testing Result of the model:

TABLE IV: TESTING RESULTS OF PROPOSED MODEL WITH OTHER

Model	Dataset	Accuracy (%)	Precision (%)	Sensitivity (%) (%)	Specificity (%)	F1-Score (%)
EfficientNetV2B3 + ViT Hybrid Model [5]	Potato Leaf Disease Dataset	85.06	82.86	85.29	84.83	83.77
CNN Model [18]	Multi-species leaf dataset	95.4	93.8	94.5	96.3	94.1
XHybrid-CNN [9]	Multi-crop plant disease dataset (15 classes)	96.41	96.4	97.0	95.82	95.7
YOLOv7 [11]	Plant disease detection dataset	90.00	91.22	87.66	92.34	89.40
Proposed Explainable Hybrid CNN	New Plant Diseases Dataset	96.90	97.85	96.80	97	96.50

A comparison of our proposed model with other models is shown in Table IV. With an accuracy of 96.90%, the proposed explainable hybrid CNN model outperforms the CNN, YOLOv7, and EfficientNet-ViT models. According to research [5], the EfficientNetV2B3 + ViT Hybrid Model for crop disease diagnosis achieves 85.06% accuracy, 82.86% precision, 85.29% Sensitivity, specificity 84.83% and 83.77% F1 score. Research using CNN [18] demonstrates 95.4% accuracy, 93.8% precision, 94.4% Sensitivity, 96.3% specificity and 94.1% F1 score. Research [9] uses the XHybrid-CNN model, which has an accuracy of 96.41%, precision of 96.4%, Sensitivity of 97%, specificity 95.82% and F1 score of 95.7%. The YOLOv7 [11] model demonstrated good plant disease detection. According to the authors of Research, they use a Plant disease detection dataset to identify crop diseases. With 90% accuracy, 91.22% precision, 87.66% Sensitivity, specificity 92.32 and F1 Score 89.40% on the plant disease detection dataset, our suggested Explainable Hybrid CNN model achieves the best results in terms of

accuracy 96.60%, precision 97.85%, sensitivity 96.80%, specificity 97%, and F1 score 89.40%. The model used in this study consistently performs well and has shown outstanding efficacy.

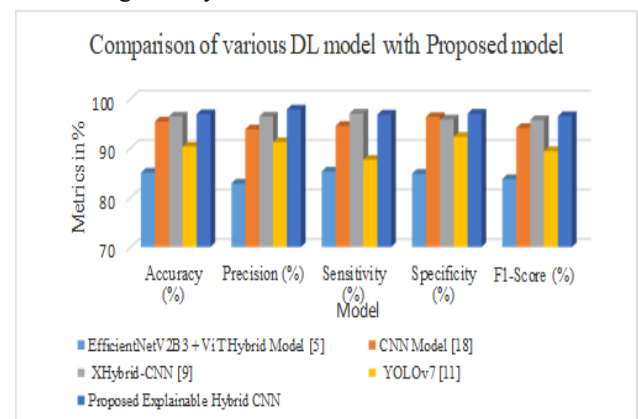


Fig. 3: Comparison of various DL model with our model

C. Predicted Result of crop leaf:

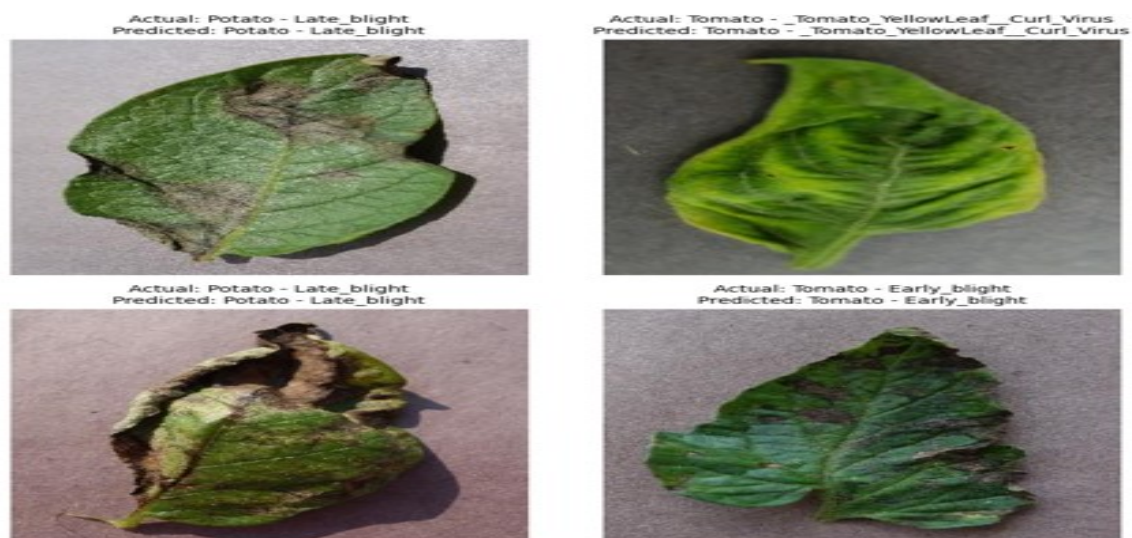


Fig. 4: Sample Predicted Images

Figure 4 depicts the outcomes of detection and classification for plant leaf images utilizing the proposed model. The input leaf is shown in each case along with the predicted disease category and confidence score. The model's capacity to accurately identify diverse plant diseases across a range of leaf shape, textures, and visual symptoms is demonstrated in the picture. In conclusion, it highlights excellent performance in real-world plant disease detection scenarios.

VI. CONCLUSION

In order to increase the detection accuracy of plant diseases and yield optimization, this research introduces an interpretable hybrid CNN framework. The suggested approach effectively provides accurate and comprehensible plant disease classification by combining interpretable AI approaches with lightweight convolutional feature extraction. The model maintains minimal computing complexity while effectively identifying disease-specific patterns across several crop species using new plant disease datasets. By providing expert validation, increasing user confidence, and clearly identifying disease-affected areas, the integration of XAI through class activation mapping improves transparency. Additionally, by calculating possible yield loss based on the features of discovered diseases, the yield optimization module extends the framework's capabilities beyond disease identification. results of experiments indicate that the proposed model performs better in terms of accuracy, precision, sensitivity, specificity, and F1-score than the conventional CNN model. The framework is perfect for real-time agricultural applications, such as mobile and edge-based deployments, due to its lightweight architecture and quick training capabilities. In summary, the suggested methodology offers a useful, dependable, and comprehensible approach to intelligent plant disease diagnosis and yield optimization, which greatly advances intelligent and sustainable agriculture. Future efforts will focus on combining multi-phase disease evolution analysis with real-time severity evaluation. The accuracy of yield forecasts can be improved by integrating IoT-based environmental sensors. Additionally, the platform might be expanded to support mobile and edge deployment for extensive field applications.

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